

Rear Wing Aerodynamics
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Table of Contents

Page

Introduction 4

Literature Review 4

CFD Simulation 5

CFD Results 7

Wind Tunnel Results 8

Comparing results 8

Experimental errors 12

Summary 12

Conclusion 13

Referencing 14

Figures

Page

Figure 1: Down force according to Bernoulli's principle (formula1-dictionary, 2023) 5

Figure 2: Numerical domain size 7

Figure 3: Pressure contour distribution around the wing at 10° AOA. 10

Figure 4: Air flow velocity streamline around the wing at 10° AOA 10

Figure 5: The Pressure contour distribution around the wing at 15° AOA 10

Figure 6: The air flow velocity streamline around the wing at 15° AOA 10

Figure 7: Pressure contour distribution around the wing at 20° AOA 11

Figure 8: The air flow velocity streamline around the wing at 20° AOA 11

Figure 9: Pressure contour distribution around the wing at 25° AOA 11

Figure 10: The air flow velocity streamline around the wing at 25° AOA 11

Figure 11: The wind tunnel numerical domain size 12

Figure 12: The wind tunnel used 13

Introduction:

The aerodynamic design of a Formula One (F1) race car is a critical aspect that engineers must consider optimising the car's performance. The rear wing of an F1 car plays a significant role in providing downforce and stabilising the car at high speeds.

In this report, we will apply air flow visualisation and CFD simulation to analyse the aerodynamic performance of the rear wing design of the F1 car. Additionally, we will compare the experimental wind tunnel test results with the CFD simulation to understand the differences in the results.

Literature Review:

Drag coefficient is a measure of how successful a streamline aerodynamic body shape is at minimising atmospheric resistance to a vehicle's acceleration.

They're numerical values produced from wind tunnel measurements or mathematical calculations (CFD). Lift tends to fall at the Critical A.O.A but drag increases dramatically. Wing stall occurs when the angle of attack exceeds this threshold.

Drag Coefficient formula: - $Cd = \frac{2 Fd}{\rho u^2 A}$

- **Cd= drag coefficient (No dimension)**
- **Fd= drag force (N)**
- **A= reference area (m2)**
- **ρ= Air density (kg/m3)**
- **u= flow speed of the object relative to the fluid (m3/s)**

The angle of attack of an aerofoil is the angle at which it flows through the air. It is calculated by dividing the chord of the aerofoil by the relative wind direction. Increasing the angle of attack deflects the airstream and creates an upward or downward pressure on the underneath or top side of the aerofoil, increasing the speed of the airflow across the aerofoil's longer side. As airflow velocity increases, pressure on the foil's top side decreases, leading to lift or downforce.

Bernoulli's Principle-The mass flow rate into the volume must equal the mass flow rate out for steady-state flow. When a pressure differential is formed, air flows at various speeds over the two sides of the wing and at various distances due to the shape of the wing. As the pressure seeks to balance, the wing attempts to move in a path of lower pressure.

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

ρ = fluid density

g = acceleration due to gravity

P_1 = pressure at elevation 1

v_1 = velocity at elevation 1

h_1 = height of elevation 1

P_2 = pressure at elevation 2

v_2 = velocity at elevation 2

h_2 = height at elevation 2

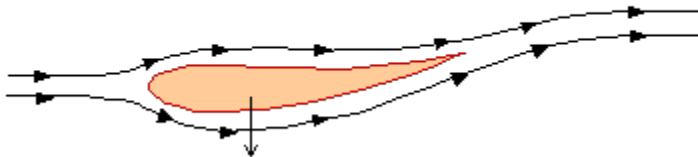


Figure 1-Down force according to Bernoulli's principle, (formula1-dictionary, 2023)

CFD Simulation:

We used SOLIDWORKS Flow Simulation to analyse the aerodynamic performance of the rear wing of the F1 car. Table 3 shows the air flow velocity streamline around the wing at different **Angles of Attack**. We analysed the wing's performance at angles of attack of 10°, 15°, 20°, and 25°. As the A.O.A increases, the downforce generated by the wing also increases, as shown in Table 1. Table 3 shows the pressure contour distribution on the wing for the same angles of attack. As expected, the low-pressure zone is concentrated on the top of the wing, while the high-pressure zone is on the

bottom of the wing. The pressure difference between the top and bottom of the wing generates the downforce that pushes the car down towards the track. We calculated the drag coefficient using the drag coefficient equation in SOLIDWORKS. The drag coefficient is a measure of the wing's ability to minimise drag while generating downforce.

The wind tunnel entrance refers to the specific point at which air enters and exits the wind tunnel during the experiment. This plays an important role in determining the conditions under which experiments are performed. In the given situation, the material used in the wind tunnel is air and its pressure is set to 101325 Pa, which is atmospheric pressure at sea level. When performing computational fluid dynamics (CFD) simulations, a numerical mesh is applied to represent the fluid body under investigation. The fluid corresponds to the flow around the rear wing. Numerical grids help discretize the computational domain, allowing fluid properties to be calculated at different locations within the domain. A boundary is created on the rear wing to define its shape and interaction with the surrounding fluid. This limitation is essential for accurately simulating the aerodynamic behaviour of wings. However, it is important to ensure that the numeric domain configuration, including boundary placement and properties, is set correctly. Using an improper numerical domain configuration can seriously affect the results of your CFD simulations. In the CFD experiment mentioned above, the region pressure and temperature are set to 101325 Pa and 293.2 K, respectively. With an air velocity of 30m/s. These values represent initial conditions in the computational domain. A pressure of 101325 Pa corresponds to atmospheric pressure and a temperature of 293.2 K represents a typical room temperature. Tuning the domain pressure and temperature is critical to create realistic initial conditions that mimic simulated real-world scenarios. These values affect flow behaviour, density, and other fluid properties within the computational domain. Well-defined initial conditions allow researchers to obtain more accurate and meaningful results from their CFD experiments. In summary, the information provided highlights the importance of the wind tunnel entrance, the impact of improper numerical domain configuration on CFD results, and the importance of setting the domain pressure and domain temperature correctly for accurate simulations. I'm here.

Numerical Domain Size

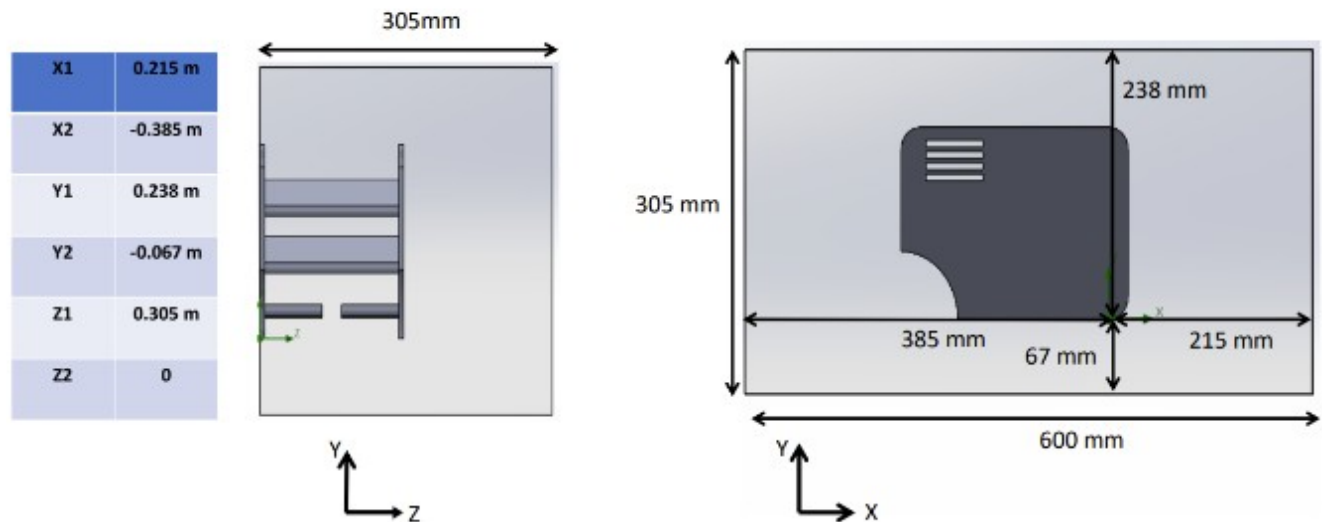


Figure 2- Numerical domain size

CFD Results:

Table 1 (CFD TEST RESULTS)-

Wing A.O.A CFD results	Drag force (N)	Downforce (N)	Drag coefficient
10°	3.895	-6.219	1.028
15°	4.750	-7.623	1.023
20°	5.406	-8.055	0.993
25°	5.879	-7.276	1.048

Table 1 shows the calculated drag coefficients for different angles of attack.

The drag increases as the angle of attack increases from 10 degrees to 25°. This is not surprising, because drag generally increases as the angle of attack increases.

The downforce values become more negative (indicating more downforce) as the angle of attack increases. This suggests that the wing can generate more downforce at higher angles of attack, improving stability and grip.

Drag coefficient is a dimensionless parameter that relates drag to other factors such as wing area and flow dynamic pressure. This provides a normalised measure of the drag experienced by the wing. The drag coefficient values differ slightly at different angles of attack, but generally remain close to 1. This indicates that the wing geometry and

design maintains consistent aerodynamic performance over the range of angles of attack tested.

Wind Tunnel Results:

Table 2 (Wind Tunnel Results)

Wing A.O.A	Drag force (N)	Downforce (N)	Drag coefficient
15°	6.17	-14.99	1.13

Table 2 shows the calculated drag coefficient for 15° A.O.A.

Drag Coefficient formula:-
$$Cd = \frac{2(6.17)}{1.18(30^2)(0.00931)}$$

$$Cd = 1.1248075799$$

The results of the wing's wind tunnel test at a set A.O.A. of 15° are shown in the table below. Drag, downforce, and drag coefficient are among the metrics that are measured. The drag force is 6.17 Newtons at a 15° A.O.A, whereas the downforce is -14.99 Newtons. The 1.13 drag coefficient indicates that, in comparison to an ideal aerodynamic shape, the drag caused by the wing design is somewhat significant. The wing generates a huge amount of downward force (-14.99 Newtons), and bears a drag force of 6.17 Newtons.

Comparing the results from CFD and wind tunnel at the same A.O.A of 15°:

1. Drag force:

- 4.750 N - CFD
- 6.17 N - Wind Tunnel

Distinction: The wind tunnel data show a larger drag force than the CFD results. This disparity could be attributed to a variety of variables, including modelling assumptions and approximations used in the CFD simulation, as well as the inherent constraints and uncertainties in wind tunnel testing.

2.Downforce:

- -7.623 N - CFD
- -14.99 N - Wind Tunnel

Distinction: The wind tunnel data show a greater magnitude of downforce than the CFD results. This discrepancy could be caused by discrepancies in modelling methodologies, experimental setups, or mistakes in either method.

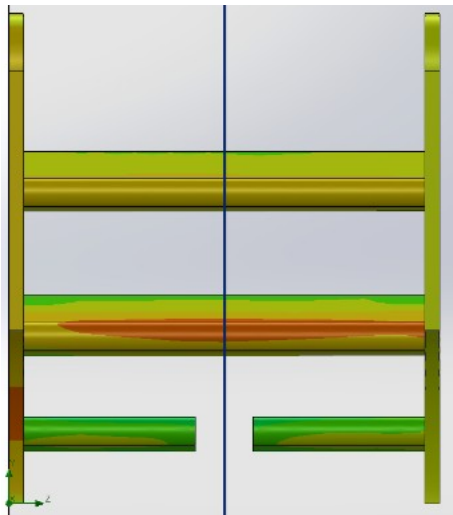
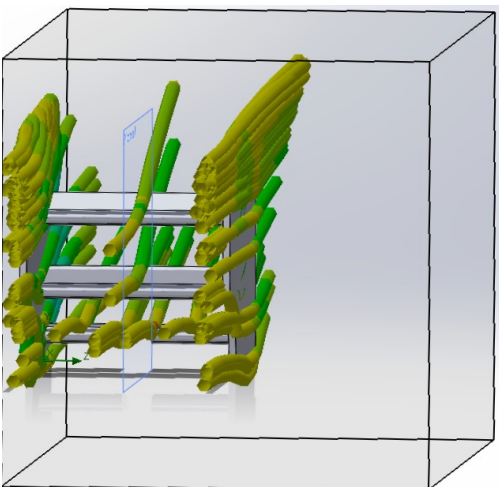
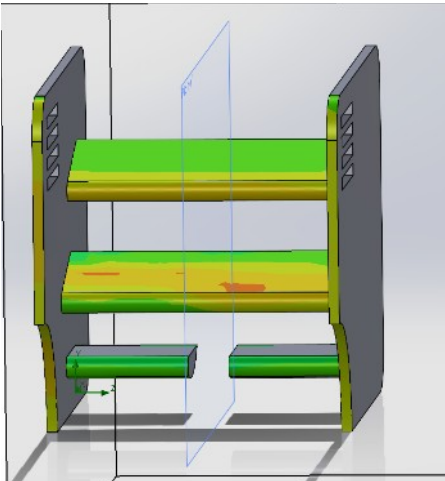
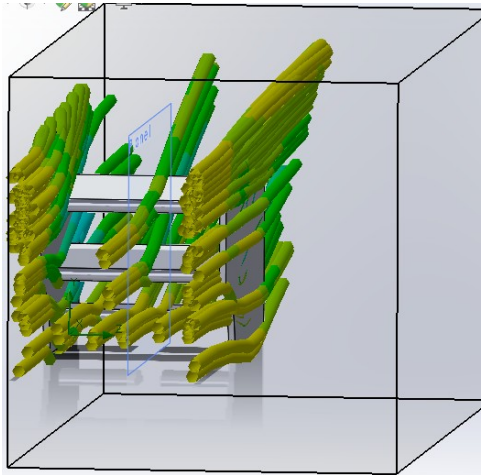
3. Drag Coefficient:

- 1.023 - CFD
- 1.13 - Wind Tunnel

Distinction: Although the drag coefficient figures are near, there is a minor difference. This disparity could be explained by the aforementioned factors influencing drag force computations.

It's crucial to remember that CFD simulations and wind tunnel testing are two independent methodologies for researching aerodynamics, each with its own set of advantages and disadvantages. Disagreements between the results achieved by these methods are not rare and are influenced by a variety of factors such as model accuracy, boundary conditions, and experimental design.

Table 3 (Different AOA showing different Pressure contour distribution and Air flow velocity)

AOA	Pressure contour distribution	Air flow velocity
10°	 Figure 3- the Pressure contour distribution around the wing at 10° AOA.	 Figure 4- the air flow velocity streamline around the wing at 10° AOA
15°	 Figure 5- the Pressure contour distribution around the wing at 15° AOA	 Figure 6- the air flow velocity streamline around the wing at 15° AOA

20°

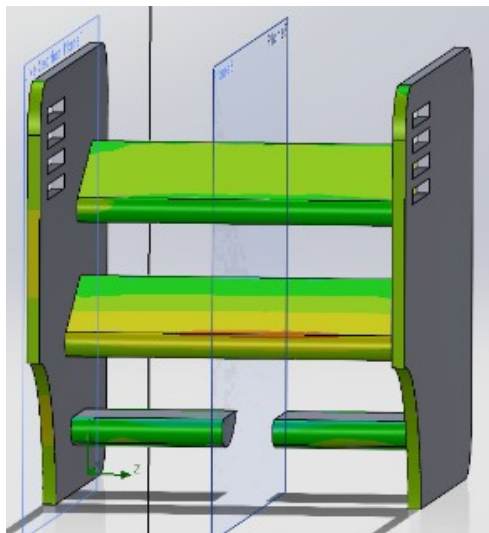


Figure 7- the **Pressure contour distribution** around the wing at **20° AOA**

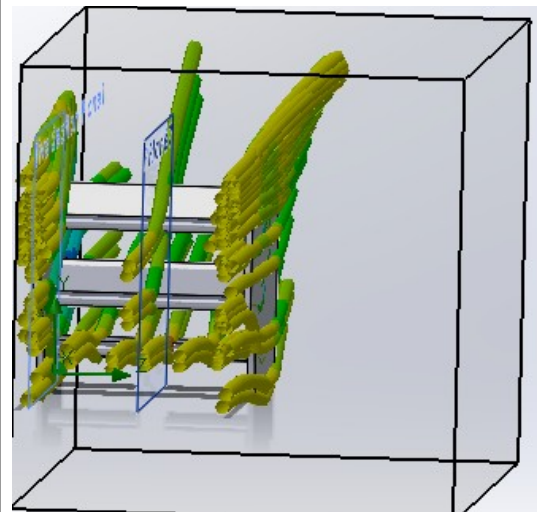


Figure 8- the air flow velocity streamline around the wing at **20° AOA**

25°

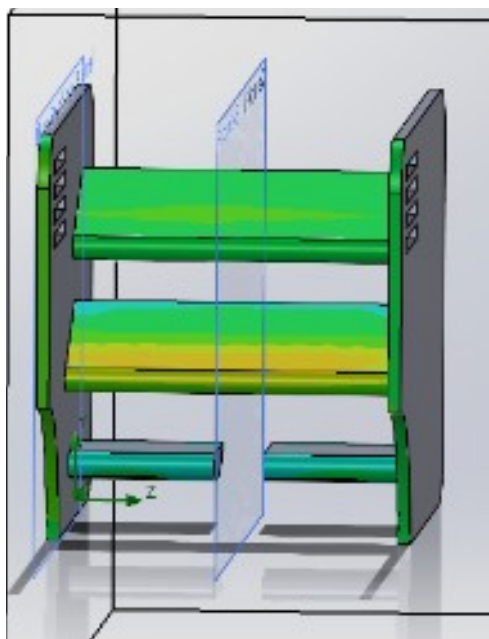


Figure 9- the **Pressure contour distribution** around the wing at **25° AOA**

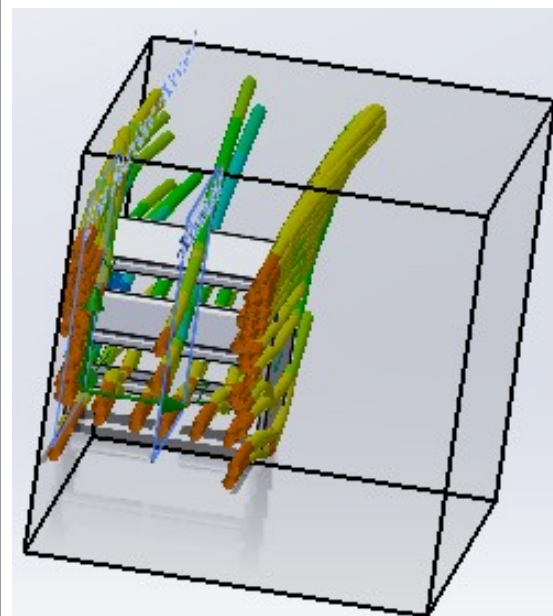


Figure 10- the air flow velocity streamline around the wing at **25° AOA**

Experimental errors:

There are several explanations why they could differ, the first being that the boundary conditions may not have been entirely precise. One reason for this might be that the wind tunnel is not smooth, resulting in a large surface gap between the surface area and air velocity. The material used to make the rear wing inside the wind tunnel was PLA. Furthermore, having an object interfere with the functioning of the wind tunnel might change the drag force, which in turn affects the drag coefficient. Another reason for two distinct outputs could be because the mesh size and numerical domain size in the CFD are slightly incorrect. The reason for this is that the CFD software is highly sensitive, and even if it is slightly inaccurate, it can still play a significant part in producing an incorrect result.

Summary: The pressure distribution results of the rear wing for different angles of attack show that increasing the angle of attack increases the downforce generated by the wing. However, increasing the angle of attack beyond a certain point may result in reduced downforce due to the separation of the airflow from the wing. The experimental wind tunnel test results showed a significant difference from the CFD simulation results. The discrepancy may be due to the complex nature of the airflow around the wing, which is challenging to replicate accurately in a wind tunnel. Additionally, experimental errors and uncertainties may have contributed to the differences in the results.

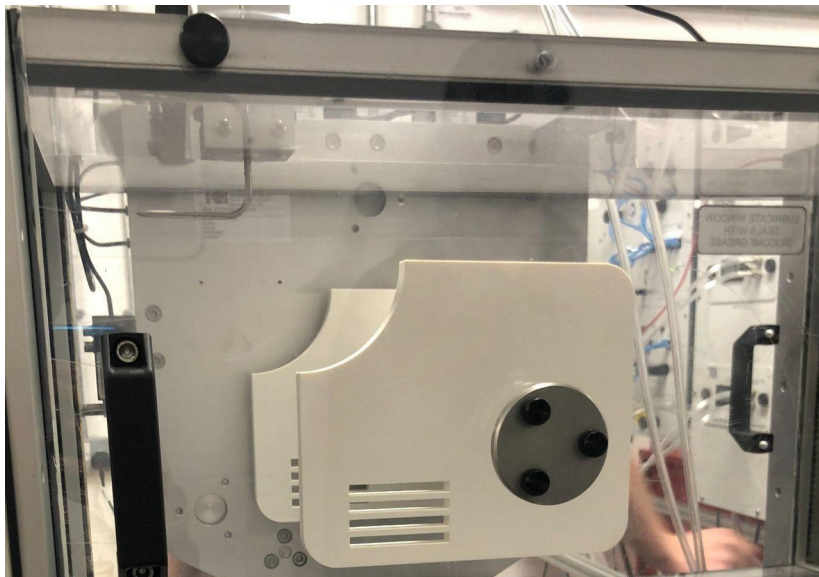


Figure 11- shows the wind tunnel numerical domain size.

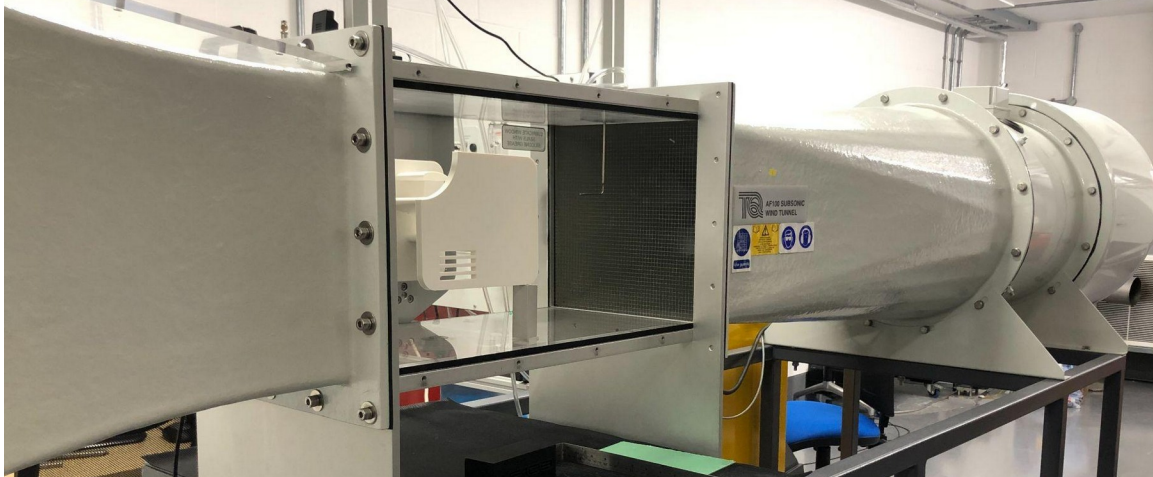


Figure 12- shows the wind tunnel used.

Conclusion:

Consider the following suggestions to improve your experiment and get more accurate results.

1. Grid improvements:

For CFD simulations, pay attention to mesh quality and grid resolution. A fine grid is used near the rear wing to accurately capture flow details. Performing a mesh sensitivity analysis by adjusting the mesh and comparing the results can ensure mesh independence and improve accuracy.

2. Wind tunnel calibration:

Calibrate the wind tunnel setup and equipment to minimize experimental error. Make sure the wind tunnel is well maintained and the flow conditions are tightly controlled. Regular wind tunnel calibration checks help maintain accuracy.

3. Material selection:

Consider using materials that are as close as possible to the real-world properties of a F1 car rear wing. The PLA mentioned in the report may have different aerodynamic properties compared to the actual wing material. Using a material with similar density, stiffness, and surface roughness to the actual wing will give more accurate results.

Implementing these improvements will increase the reliability and accuracy of our experiments, allowing us to draw more informed conclusions about the aerodynamic performance of rear wings in Formula 1 cars.

Referencing-

Figure 1-

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